

App Development Project Report

TREKISTA- trek management app

1. Student Details

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About Me: I am a student at IIT Madras BS Degree program with an interest in web application development and data-driven technologies. I enjoy learning about meaningful technology and applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Trekista – trek management app

Problem Statement:

To design and build a web-based application that allows users to **explore, plan, book treks, and**, plan their adventures. A super admin handles the management and trek staff that looks over users, treks and bookings.

Approach:

The app was built using Flask as the backend framework with a modular structure. It is made on MVC architecture. Three roles are defined in it- admin, trek staff, user.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, and improving variable naming consistency, for a building guideline for step-by-step building path.

The extent of AI/LLM usage is around **15-20%**, limited to **code suggestions, syntax of function usage, documentation formatting**, and minor debugging.

Almost all final implementation logic, routing and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
HTML – CSS	For frontend styling and web page design
SQLite	Lightweight local database for storing user data

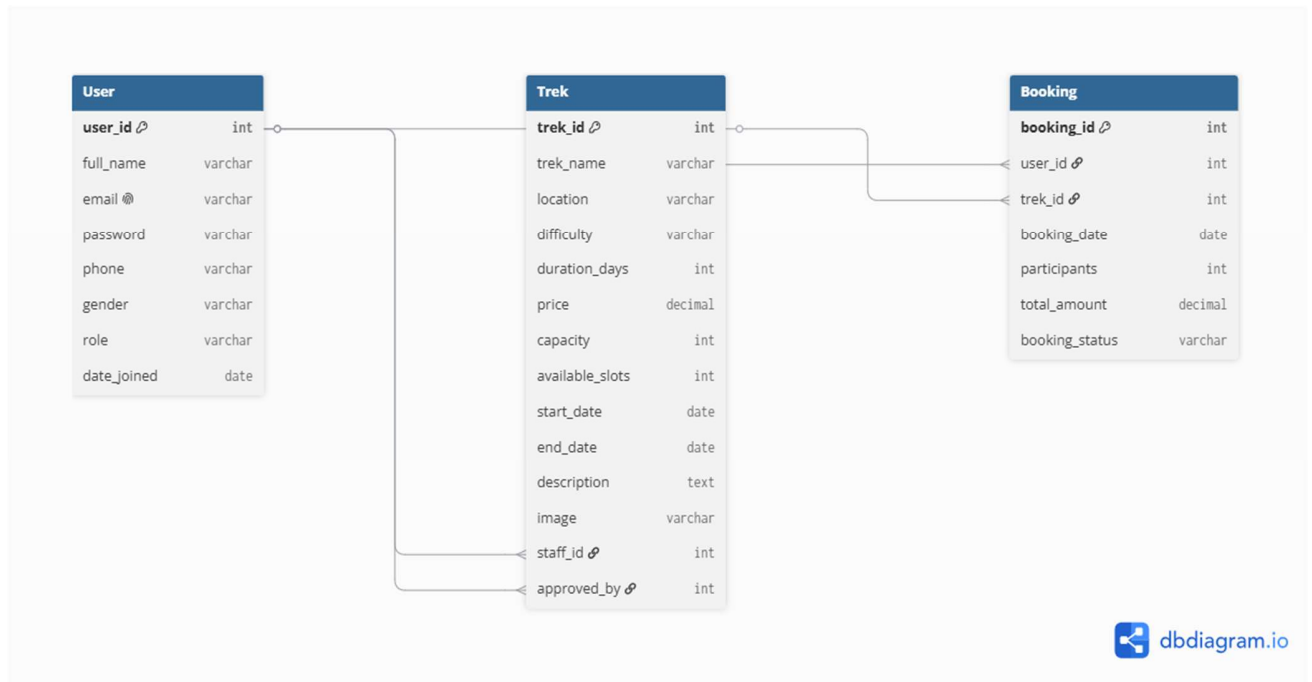
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user details (id, name, email, password, role)
2. **profile**— logs user and details (id, date, user info)
3. **bookings**— logs user bookings (id, user_id, trek_id, duration, date)
4. **treks**— stores trek details(id, trek_name, difficulty,)

Relationships:

- One-to-Many → **User** → **bookings**
- One-to-Many → **trek** → **users**



(madewith dbdiagram.io)

6. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for user and routes
- **/templates** – Jinja2 HTML templates
- **/controllers** – handling the routing and functionalities
- **/static** – CSS, and upload images

Implemented Features:

1. **Admin:** Admin is the pre-existing superuser of the application.
 - Can add, edit, or remove trek
 - Can approve and blacklist trek staff
 - Can assign staff to a particular trek
 - Can view all the users, staff, and treks
 - Can search treks, staff, or users by name or ID

- Can blacklist users or staff if required
- 2. Trek Staff:** staff members that carry out trekking and guide trekkers
- Can register and log in
 - Need admin's approval to access the dashboard
 - Can view assigned treks by Admin
 - Can update trek details such as:
 - Available slots
 - Trek status (Open/Closed)
 - Can view list of users registered for their treks
- 3. User (Trekker)**
- Can register and log in
 - Can view approved/open treks
 - Can search and filter treks based on difficulty, location
 - Can book treks
 - Can view booking status and trekking history
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7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1tA5WphvIZl3nH8IE17VIM7tbO47chh3j/view?usp=sharing>

(Accessible to all.)